

# **A Project Report on Skill Exchange System**

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## **SUPERVISOR RECOMENDATION**

This is to certify that this project entitled, “**Skill Exchange System**” prepared and submitted by **Anubhav Dahal, Sulav Raj Shrestha, Grishma Adhikari** and **Krish Shrestha** in partial fulfillment of the requirements of the degree of Bachelor of Computer System and Information Technology (BCSIT) awarded by Pokhara University, has been completed under my supervision. I recommend the same for acceptance by the University.

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## CERTIFICATE OF APPROVAL

This project entitled, “**Skill Exchange System**” prepared and submitted **Anubhav Dahal, Sulav Raj Shrestha, Grishma Adhikari** and **Krish Shrestha**, has been examined and is accepted for the award of the degree of Bachelor of Computer System and Information Technology (BCSIT) awarded by Pokhara University.

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## **ABSTRACT**

Skill Exchange System which is called SkillSwap, is a web-based platform designed to connect individuals who require services with skilled individuals who can provide them. In many communities, people face difficulties in finding reliable service providers for tasks such as tutoring, technical support, repairs, and other everyday services. At the same time, many skilled individuals lack a proper platform to showcase their abilities and reach potential customers.

The main objective of this project is to develop a centralized and user-friendly system that facilitates skill sharing and service exchange within the community. The platform allows users to register, create profiles, search for available services, and connect with skilled workers based on their requirements. The system also supports service management, booking requests, and secure user authentication.

The Skill Swap platform was developed using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL. Various software engineering methodologies were applied during requirement analysis, system design, implementation, and testing phases to ensure the reliability and usability of the system.

The proposed system aims to improve accessibility to local services, promote community collaboration, and provide opportunities for skilled individuals to gain recognition and employment. Future enhancements may include mobile application development, AI-based service recommendations, enhanced user verification, and integration of secure online payment gateways to further improve the platform's functionality, accessibility, and overall user experience.

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## ABBREVIATIONS

| Abbreviation | Meaning                            |
|--------------|------------------------------------|
| API          | Application Programming Interface  |
| CSS          | Cascading Style Sheets             |
| DBMS         | Database Management System         |
| ER           | Entity Relationship                |
| GPS          | Global Positioning System          |
| HTML         | HyperText Markup Language          |
| HTTP         | HyperText Transfer Protocol        |
| IDE          | Integrated Development Environment |
| JS           | JavaScript                         |
| JSON         | JavaScript Object Notation         |
| OTP          | One-Time Password                  |
| P2P          | Peer-to-Peer                       |
| PostgreSQL   | Postgre Structured Query Language  |
| SQL          | Structured Query Language          |
| UI           | User Interface                     |
| URL          | Uniform Resource Locator           |
| UX           | User Experience                    |
| VS Code      | Visual Studio Code                 |



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# CHAPTER 1: INTRODUCTION

## 1.1. Background of the Study

With the rapid growth of digital technology, online platforms have become an effective medium for connecting individuals and providing services. Many people possess valuable skills and expertise that remain underutilized, while others often struggle to find reliable assistance for small tasks and services. Traditional methods of finding skilled individuals are often time-consuming, expensive, and lack transparency.

SkillSwap (SS) is a web-based application developed to bridge the gap between people who need assistance and those who can offer their skills. The system provides a centralized platform where users can showcase their abilities, search for services, communicate with others, and request assistance. The platform aims to promote community-based skill sharing by enabling direct interaction between service seekers and service providers.

The project is based on the concepts of the sharing economy and peer-to-peer service exchange, where users can connect directly without relying on intermediaries. By creating a trusted environment with user profiles, ratings, reviews, and communication features, Skill Swap encourages collaboration and efficient utilization of community skills.

## 1.2. Statement of the Problem

Many individuals face difficulties in finding trustworthy and skilled people for small tasks such as tutoring, repairs, technical assistance, and other everyday services. Existing solutions often focus on paid services or online learning platforms and provide limited support for local community-based interactions.

The major problems identified are:

- i. **Lack of a Centralized Platform:** There is an absence of a unified, localized digital directory dedicated to immediate, task-oriented skill exchanges, forcing individuals to rely on fragmented, inefficient communication channels like social media groups or word-of-mouth.
- ii. **Information Asymmetry and Trust Deficit:** Engaging unknown independent service providers introduces significant risks regarding reliability, quality, and

personal safety. Without a transparent, historical verification system, community trust remains compromised.

- iii. **Economic Inefficiencies:** For minor, non-professional tasks, mainstream commercial service providers charge premium rates driven by administrative overheads and middleman interventions, making simple tasks financially burdensome for demographic segments like university students.
- iv. **Underutilization of Local Talent:** Competencies possessed by community members frequently remain dormant or unmonitored, limiting opportunities for experiential learning, peer-to-peer mentorship, and mutual community support.

Therefore, there is a need for a system that can efficiently connect people, promote trust and transparency, and facilitate skill sharing within communities.

### 1.3. Objectives of the Study

The main objective of the project is to develop a web-based Skill Swap platform that connects individuals seeking assistance with those willing to offer their skills.

The specific objectives are:

- i. To Connect Customers with Skilled Workers:
- ii. To Promote Skill Sharing in the Community:
- iii. To Make Service Booking Easy and Fast:
- iv. To Build Trust Through Ratings and Reviews:
- v. To Manage Offline Services Efficiently:

### 1.4. Scope of the Study

The primary focus of the SkillSwap is to develop a secure, user-friendly, and community-oriented web platform that connects individuals who want to offer skills with those seeking services. The scope of the system includes:

- i. **User Registration and Authentication:** Secure user registration, login, and logout with input validation and unique email verification.
- ii. **Profile Management:** Users can manage their personal information, update their profile, and choose to act as either a Skill Seeker or a Skill Provider.
- iii. **Skill Management:** Skill Providers can create and manage service listings by adding skill details, categories, descriptions, and service charges.

- iv. **Search and Discovery:** Users can search and browse available skills using keywords and location filters.
- v. **Booking Management:** Users can request services, while providers can accept or reject booking requests.
- vi. **Messaging System:** Registered users can communicate securely through the built-in real-time messaging feature before providing the service.
- vii. **Rating and Review System:** Users can provide ratings and reviews after completing a service to help maintain service quality and trust.
- viii. **Community-Based Service Exchange:** The system supports connecting users for **physical, face-to-face service exchanges** within the community after successful communication through the platform.

### 1.5. Limitations of the Study

Although the SkillSwap platform provides a practical solution for connecting Skill Seekers and Skill Providers, it has the following limitations:

- i. **Web-Based Platform:** The system is developed as a web application and does not currently provide a dedicated mobile application for Android or iOS.
- ii. **User Verification:** The platform does not integrate with third-party identity verification services. User information and listed skills are based on the details provided during registration.
- iii. **Location Support:** The system uses user-provided location information and does not support real-time GPS tracking or distance-based recommendations.
- iv. **Limited Administration Features:** While administrators can manage users and payments, advanced features such as automated content moderation and dispute resolution are not currently implemented.
- v. **Internet Dependency:** The system requires an active internet connection to access its features, as offline functionality is not supported.

## **CHAPTER 2: REVIEW OF LITERATURE**

### **2.1. Description of Fundamental Theories, General Concepts and Terminologies**

The Skill Swap platform is developed based on the concepts of the Sharing Economy and Peer-to-Peer (P2P) Service Exchange Systems. The sharing economy refers to an economic model where individuals share resources, skills, and services directly with one another through digital platforms rather than relying on traditional organizations or intermediaries. This approach promotes efficient utilization of available skills within a community while reducing costs and increasing accessibility to services.

Similarly, the peer-to-peer model enables direct interaction between users who need services and users who are willing to provide them. Instead of depending on third-party agencies, individuals can communicate, exchange information, and arrange service requests through a centralized online platform. This concept forms the core foundation of the Skill Swap system.

The platform incorporates several important concepts and terminologies that are essential for its operation:

- User Roles:

The system allows users to perform multiple roles based on their needs. A user can act as a Skill Seeker, who requests assistance for a particular task, or as a Skill Provider, who offers a skill or service to others. This flexibility encourages active participation and promotes mutual skill sharing within the community.

- Session Management:

Session management is a backend mechanism used to maintain and monitor authenticated user activities throughout their interaction with the system. Once a user logs into the platform, a secure session is created that allows the user to access protected features without repeatedly entering login credentials. This improves both security and user experience.

- Booking and Request Management System:

The booking and request management system handles the complete workflow of service exchanges. Users can submit requests for services, which are then received by service providers. Providers can accept, reject, or manage requests according to their availability. The system tracks the status of each request from creation to completion, ensuring proper record keeping and transparency.

- **Communication System:**

Effective communication is a key component of the Skill Swap platform. The system provides messaging and interaction features that enable seekers and providers to discuss service details, schedules, expectations, and other relevant information before confirming a service exchange. This helps reduce misunderstandings and improves coordination between users.

- **Rating and Review System:**

To establish trust and credibility within the platform, a rating and review mechanism is incorporated. After completing a service exchange, users can provide ratings and feedback based on their experience. These ratings help future users evaluate the reliability and quality of service providers, creating a transparent and trustworthy environment.

- **Community-Based Service Exchange:**

Unlike many existing platforms that focus primarily on paid services or online learning, Skill Swap emphasizes community-driven collaboration. The platform encourages local interaction, skill utilization, and mutual support among community members, helping individuals connect with trusted people in their surrounding areas.

By integrating these concepts and technologies, the Skill Swap platform provides a reliable, efficient, and user-friendly environment for connecting skilled individuals with people seeking assistance, ultimately promoting collaboration and community development.

## **2.2. Review of the similar Projects and Theories**

Several existing skill-sharing and online learning platforms were studied to understand their features, strengths, and limitations. Platforms such as Skillshare, Skillpact, and

Skilljar provide online environments where users can learn new skills through courses, training materials, and collaborative activities.

These systems commonly offer features such as:

- User registration and authentication
- Skill or course listing
- Communication and collaboration tools
- Rating and review mechanisms
- Progress tracking and learning resources

Despite their usefulness, these platforms have certain limitations:

- They primarily focus on online learning and educational content.
- Opportunities for local community interaction are limited.
- Support for face-to-face skill exchange and offline collaboration is minimal.
- Real-world peer-to-peer service exchange is often unavailable.
- Many platforms emphasize paid courses rather than community-based knowledge sharing.

The SkillSwap system extends beyond traditional online learning platforms by creating a community-oriented environment where users can both learn and provide skills. Unlike existing systems that mainly focus on educational courses, SkillSwap supports a wider range of skills and services such as technology, design, language learning, tutoring, home maintenance, photography, cooking, arts, and other community-based activities.

The proposed system offers several advantages:

- Supports offline skill exchange which results in better efficiency.
- Encourages local community engagement and collaboration.
- Allows direct interaction between skill providers and skill seekers.
- Facilitates peer-to-peer service exchange rather than only course-based learning.



- Provides service booking and scheduling functionality.
- Creates opportunities for practical, real-world learning experiences.
- Promotes knowledge sharing within local communities.
- Enables users to build reputation through ratings and reviews.

Several research studies related to peer-to-peer learning and skill exchange were also reviewed during the requirement identification phase.

A study by Sri (2025) emphasized the importance of user profiles, communication systems, and collaborative learning in skill-sharing applications. Similarly, Jadhav et al. (2025) highlighted matchmaking mechanisms and community interaction as key components of successful peer-to-peer skill exchange platforms.

Anam et al. (2025) proposed a web-based collaborative learning platform developed using PHP and MySQL, providing features such as authentication, user management, and skill administration. Furthermore, Ukata and Moko (2026) discussed the effectiveness of peer-to-peer interaction and practical learning through mobile-based skill exchange applications.

These studies helped identify important system requirements such as communication, booking management, user profiles, and community-based interaction for the proposed system.

## **CHAPTER 3: SYSTEM ANALYSIS AND DESIGN**

### **3.1. System Analysis**

System analysis is an important phase of the Software Development Life Cycle (SDLC) that focuses on understanding the problems, requirements, and objectives of a system before development begins. It involves studying user needs, identifying system functionalities, and determining how the proposed system will solve existing problems.

For the Skill Swap system, system analysis was conducted to understand the requirements of service seekers, service providers, and administrators. This process helped identify the necessary features, data requirements, system processes, and operational constraints. It also included evaluating the feasibility of the system, analyzing data flow, and designing solutions that ensure efficient skill sharing and service exchange within the community.

#### **3.1.1. Requirement Analysis**

##### **3.1.1.1. Function Requirements:**

Functional requirements describe the specific operations and services that the Skill Swap system must perform. These requirements are represented in the Use Case Diagram, which illustrates the interactions between different actors and the system.



**ii. Profile Management:**

Users shall be able to update personal information, profile pictures, skill details, contact information, and account preferences.

**iii. Skill Management:**

Skill Providers shall be able to add, edit, update, and remove skills offered through the platform. Providers may also specify service descriptions, pricing information, and availability.

**iv. Search and Service Discovery:**

Skill Seekers shall be able to search and browse available skills and services using keywords to find suitable providers.

**v. Booking and Scheduling:**

The system shall allow seekers to send booking requests and select preferred dates and times. Providers shall have the ability to accept, reject, or manage incoming requests.

**vi. Communication System:**

The platform shall provide messaging functionality that enables direct communication between service seekers and service providers before and after booking confirmation.

**vii. Payment Management:**

The system shall support secure payment processing between seekers and providers while maintaining records of completed transactions.

**viii. Rating and Review System:**

After completing a service, users shall be able to submit ratings and reviews that contribute to the trustworthiness and reputation of service providers.

**ix. Report Management:**

Users shall be able to report inappropriate behavior, fraudulent activities, or policy violations. Administrators shall review and resolve reported issues.

**3.1.1.2. Non-Functional Requirements:**

Non-functional requirements define the quality attributes, performance expectations, and operational constraints of the Skill Swap system.

**i. Performance Requirements:**

The system shall provide fast response times for user interactions such as login, searching, booking, and messaging. Most user requests should be processed within a few seconds under normal operating conditions.

## **ii. Security Requirements:**

User passwords shall be securely stored using encryption and hashing techniques. Authentication and session management mechanisms shall protect user accounts from unauthorized access. Sensitive information such as payment data shall be handled securely.

## **iii. Reliability Requirements:**

The system shall maintain stable operation and provide consistent access to services with minimal downtime. Data integrity shall be preserved during all transactions and system operations.

## **iv. Scalability Requirements:**

The system shall support future growth in the number of users, skills, bookings, and transactions without significant performance degradation.

## **v. Usability Requirements:**

The user interface shall be simple, intuitive, and easy to navigate. Users with basic computer knowledge should be able to access system features without extensive training.

## **vi. Compatibility Requirements:**

The web application shall be accessible through modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The interface shall adapt to desktop, tablet, and mobile devices.

## **vii. Maintainability Requirements:**

The system shall be developed using a modular structure, allowing future updates, bug fixes, and feature enhancements to be implemented efficiently.

## **viii. Availability Requirements:**

The platform shall be available to users whenever internet connectivity is available, ensuring continuous access to services and user data.

### **3.1.2. Feasibility Analysis**

#### **3.1.2.1. Technical Feasibility**

The proposed system is technically feasible because it is developed using widely available web technologies and can be deployed on cloud hosting platforms.

The technologies used for system development include:

- **Frontend:** HTML, CSS, JavaScript

- **Backend:** Node.js with Express.js
- **Database:** MySQL (development) and PostgreSQL (deployment on Render)
- **Hosting Platform:** Render

The required tools, frameworks, and development environments are accessible and suitable for a student-level project. The development team has basic knowledge of these technologies, making implementation achievable.

#### **3.1.2.2.Operational Feasibility**

The system is operationally feasible because it is designed to be simple and user friendly.

Users can easily:

- Register and create profiles
- Search for services
- Communicate with providers
- Book services
- Make payments and provide feedback

The platform supports practical real-world interaction and addresses the problem of finding trusted local services efficiently.

#### **3.1.2.3.Financial Feasibility**

The project is economically feasible because the development cost is relatively low. Most of the required technologies and tools are open-source and freely available. The project mainly requires:

- Internet access
- Development tools
- Basic hosting services

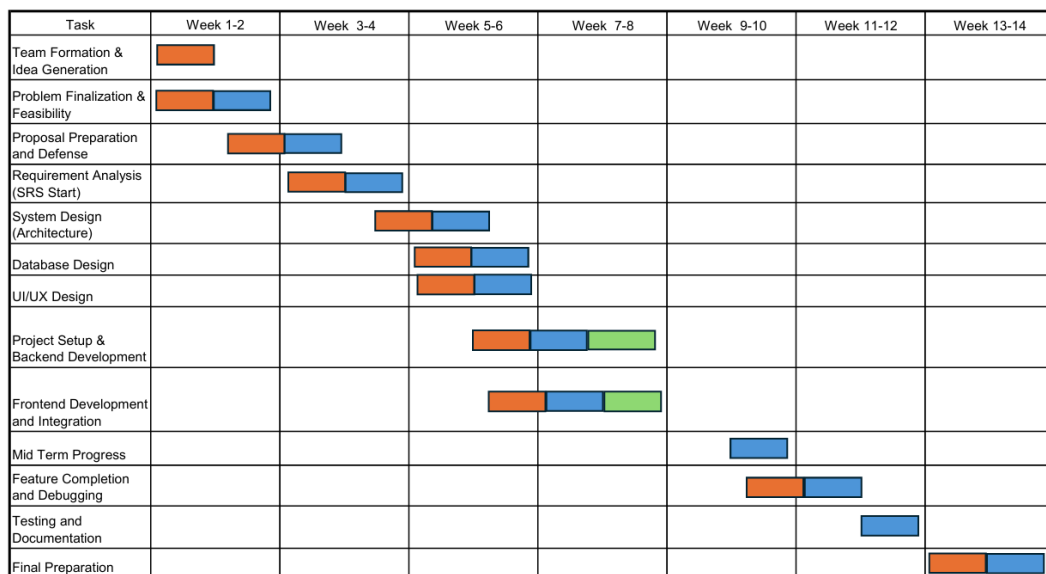
Since the system is developed as an academic project, no major financial investment is required. Additionally, the platform can generate revenue in the future through commission-based payments from service transactions.

#### **3.1.2.4.Schedule Feasibility**

Schedule feasibility evaluates whether the proposed system can be developed and implemented within the available time frame. The development of the Skill Swap system was planned according to the academic project schedule and divided into several phases, including requirement analysis, system design, database design, interface development, backend development, testing, and documentation.

The project was considered schedule feasible because the required technologies such as HTML, CSS, JavaScript, Node.js, and MySQL are familiar to the development team and the project scope is manageable within the allocated semester duration. Proper planning and task distribution among team members helped ensure that each development phase could be completed on time.

Therefore, the Skill Swap system can be successfully developed, tested, and documented within the given project schedule without requiring additional time or resources.



**Figure 2: Gantt Chart**

### 3.1.3. Data Modeling

Data modeling describes the data required by the system and the relationships between different entities.

#### Main Entities

- User

- Service
- Booking
- Payment
- Review
- Admin

### **Relationships**

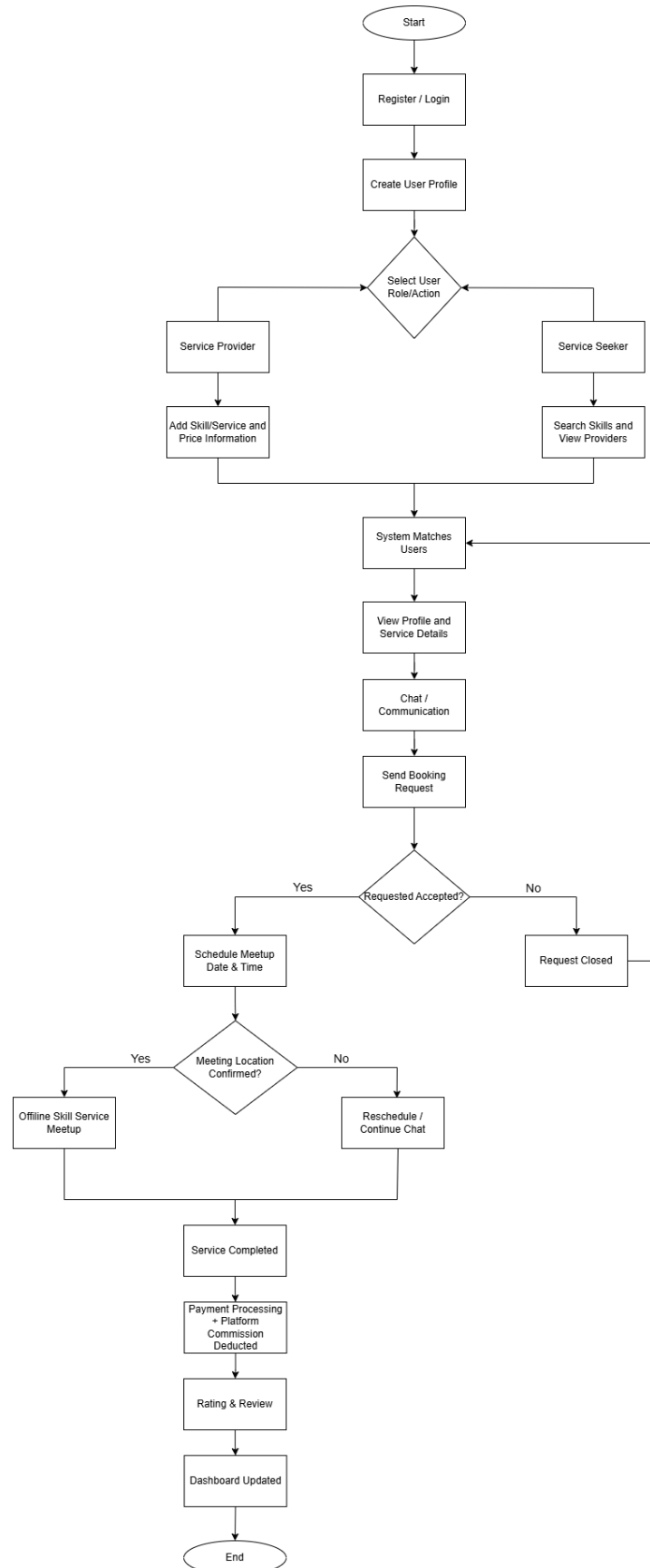
- One User can create many Services.
- One Service belongs to one Service Provider.
- One Service can have many Bookings.
- One Booking belongs to one Service Seeker.
- One Booking generates one Payment.
- One Booking can receive one Review.
- Admin manages users, services, bookings, and payments.

The system stores user information, service details, booking records, payment information, and reviews. These entities are connected through primary and foreign keys to ensure data integrity and consistency.

#### **3.1.4. Process Modeling**

Process modeling describes how data flows through the system and how users interact with the platform.





**Figure 3: Flowchart**

The system flowchart represents the overall workflow of the platform from user registration to service completion.

The process begins when users register and create profiles. Service providers add skills and pricing information, while service seekers search for services and providers. The system matches users based on their requirements and enables communication through the chat system.

After communication, booking requests are sent and managed by the provider. Once accepted, users schedule an offline meetup to complete the service. After successful completion, payment is processed, platform commission is deducted, and ratings and reviews are provided.

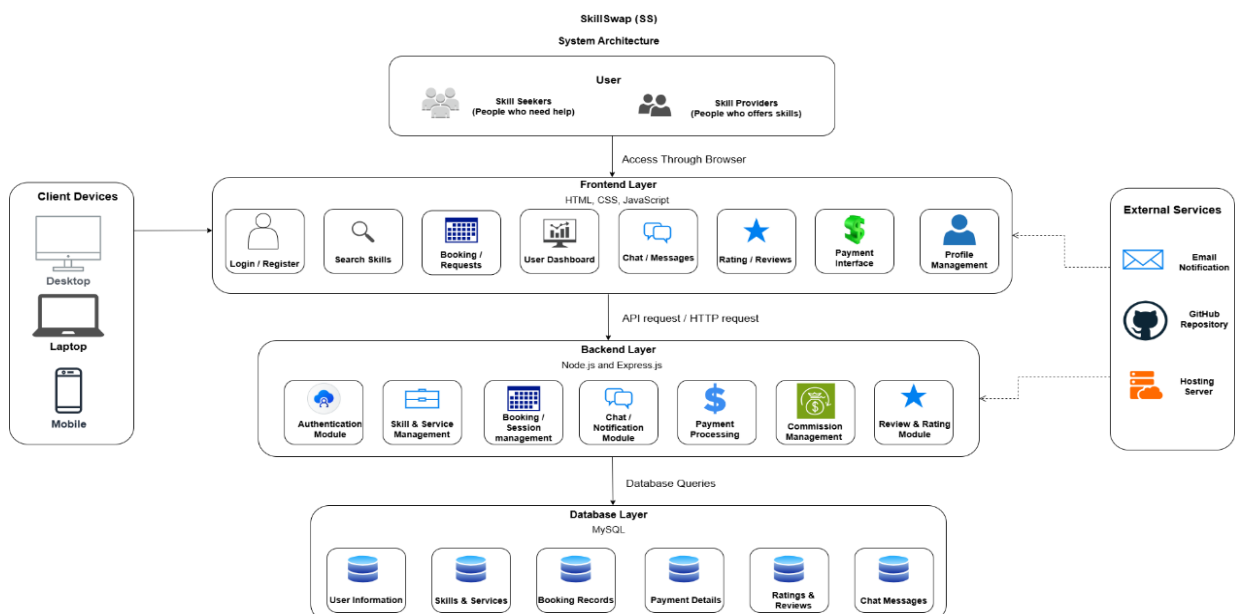
The flowchart helps visualize the step-by-step operation of the entire system.

## 3.2. System Design

The SkillSwap platform follows a three-tier architecture consisting of:

- Presentation Layer (Frontend)
- Application Layer (Backend)
- Data Layer (Database)

### 3.2.1. Architectural Design



**Figure 4: System Architecture**

### **i. Presentation Layer (Frontend)**

This layer handles user interaction and interface design. Users interact with the system through web pages developed using:

- HTML
- CSS
- JavaScript

This layer includes:

- Login pages
- Dashboard
- Skill listing pages
- Booking interfaces
- Chat interfaces

### **ii. Application Layer (Backend)**

The backend processes business logic and system operations. It manages:

- User authentication
- Booking management
- Chat handling
- Payment processing
- Rating and review management

The backend will be developed using:

- Node.js
- Express.js

### **iii. Database Layer**

This layer stores and manages all system data including:

- User information

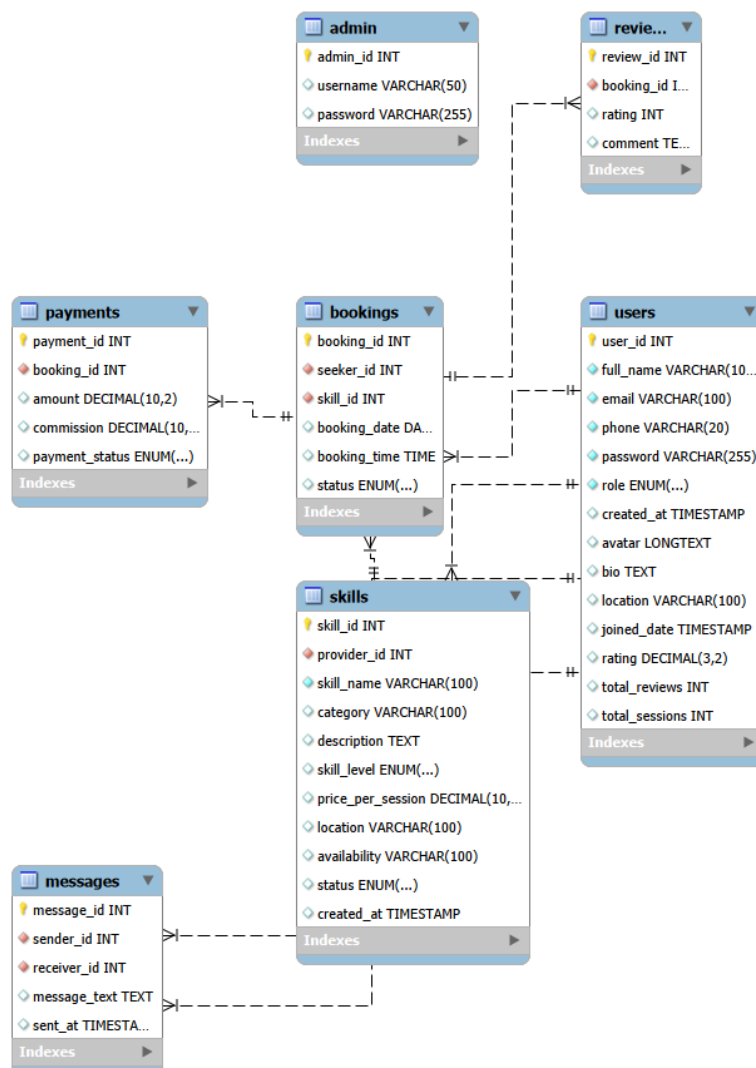
- Skills and services
- Bookings
- Payments
- Ratings and reviews

The database system used is:

- MySQL

### 3.2.2. Database Schema Design

The database schema of the SkillSwap platform was designed using MySQL Workbench. The schema consists of several entities including User, Service, Booking, Payment, Review, and Admin. These entities are interconnected through primary and foreign key relationships to ensure data integrity and efficient data management.



**Figure 5: ER Diagram**

### **3.2.3. Interface Design**

### **3.2.4. Physical Model**

#### **Hardware Requirements**

- Processor: Intel i3 or higher
- RAM: 4GB minimum
- Storage: 20GB free space
- Internet Connection

#### **Software Requirements**

- Windows 10/11
- VS Code
- Node.js
- MySQL Workbench
- Web Browser (Chrome, Firefox, Microsoft Edge)

## **CHAPTER 4: IMPLEMENTATION AND TESTING**

### **4.1. Implementation**

Implementation involves converting the system design into a working software application.

The SkillSwap platform was developed using Node.js, Express.js, MySQL, HTML, CSS, and JavaScript.

#### **4.1.1. Tools Used**

**Table 1: Tools Used**

| <b>Category</b>      | <b>Tool</b> |
|----------------------|-------------|
| CASE TOOL            | Figma       |
| Programming Language | JavaScript  |
| Frontend             | HTML, CSS   |

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| Backend         | Node.js, Express.js                                       |
| Database        | MySQL (development) and PostgreSQL (deployment on Render) |
| IDE             | VS Code                                                   |
| API Testing     | Postman                                                   |
| Version Control | GitHub                                                    |

#### **4.1.2. Implementation Details of Modules**

##### **User Management Module**

Functions:

- User Registration
- Login
- Logout
- Profile Management

##### **Service Management Module**

Functions:

- Create Service
- Edit Service
- Delete Service
- Search Services

##### **Booking Module**

Functions:

- Create Booking
- Cancel Booking
- View Booking Status

## **Payment Module**

Functions:

- Process Payments
- Calculate Commission
- Store Transaction Records

## **Review Module**

Functions:

- Submit Review
- View Reviews
- Rating Calculation

## **Admin Module**

Functions:

- Manage Users
- Manage Services
- Monitor Payments
- Generate Reports

## **4.2. Testing**

Testing was performed to verify that all modules function correctly and satisfy user requirements.

#### 4.2.1. Test Cases for Unit Testing

**Table 2: Test Cases for Unit Testing**

| Test Case ID | Module           | Test Scenario                    | Expected Result                  | Status |
|--------------|------------------|----------------------------------|----------------------------------|--------|
| UT-01        | Registration     | Enter valid user details         | Account created successfully     | Pass   |
| UT-02        | Registration     | Leave full name empty            | Validation message displayed     | Pass   |
| UT-03        | Registration     | Enter invalid email              | Email validation error displayed | Pass   |
| UT-04        | Login            | Enter correct email and password | User logged in successfully      | Pass   |
| UT-05        | Login            | Enter incorrect password         | Login denied with error message  | Pass   |
| UT-06        | Profile          | Update profile information       | Profile updated successfully     | Pass   |
| UT-07        | Service Creation | Add a new skill/service          | Skill saved in database          | Pass   |
| UT-08        | Booking          | Book an available service        | Booking created successfully     | Pass   |
| UT-09        | Messaging        | Send message to another user     | Message delivered successfully   | Pass   |



|       |         |                              |                               |      |
|-------|---------|------------------------------|-------------------------------|------|
| UT-10 | Payment | Complete payment for booking | Payment recorded successfully | Pass |
|-------|---------|------------------------------|-------------------------------|------|

#### 4.2.2. Test Cases for System Testing

**Table 3: Test Cases for System Testing**

| Test Case ID | System Function              | Expected Result                                              | Status |
|--------------|------------------------------|--------------------------------------------------------------|--------|
| ST-01        | User Registration<br>→ Login | User can login after successful registration                 | Pass   |
| ST-02        | Login → Home Page            | Profile menu displayed after login                           | Pass   |
| ST-03        | Logout                       | User session ends and Join Now button is displayed           | Pass   |
| ST-04        | Unauthorized Access          | Protected pages redirect unauthenticated users to login page | Pass   |
| ST-05        | Profile Management           | Updated profile information displayed correctly              | Pass   |
| ST-06        | Skill Posting                | Newly added skills appear on Find Skills page                | Pass   |

|       |                    |                                          |      |
|-------|--------------------|------------------------------------------|------|
| ST-07 | Booking Workflow   | User can successfully book a service     | Pass |
| ST-08 | Messaging System   | Users can exchange messages in real time | Pass |
| ST-09 | Payment Processing | Payment information stored successfully  | Pass |

## CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATION

### 5.1. Lesson Learnt

The development of the **Skill Swap** platform provided valuable practical experience in software development and project management. Through this project, the team gained knowledge in requirement analysis, system design, database modeling, frontend and backend development, deployment, and software testing.

The project enhanced our understanding of web technologies including HTML, CSS, JavaScript, Node.js, Express.js, MySQL, and PostgreSQL. We learned how different components of a web application work together to deliver a secure and user-friendly platform. We also gained hands-on experience in deploying the application using Render, which provided a free cloud hosting environment for our backend and PostgreSQL database.

Furthermore, the project strengthened our understanding of software engineering concepts such as system analysis, database design, testing methodologies, deployment, and technical documentation. It also improved our teamwork, communication, problem-solving, and time-management skills while demonstrating the importance of developing solutions that effectively address real-world community needs.

## **5.2. Conclusion**

The Skill Swap platform was developed to provide a centralized environment where individuals can connect with skilled service providers and access various services conveniently. The system addresses the challenges of finding reliable assistance, promotes community-based skill sharing, and creates opportunities for individuals to showcase their talents and expertise.

The developed platform successfully implements core functionalities including user registration, authentication, profile management, service management, service discovery, and booking management. The system was tested to ensure that its major functions operate correctly and meet the specified requirements.

Overall, the project achieved its primary objective of creating a web-based platform that facilitates interaction between service seekers and service providers. The Skill Swap platform has the potential to improve community collaboration, increase access to local services, and promote the effective utilization of available skills within society.

## **5.3. Recommendations**

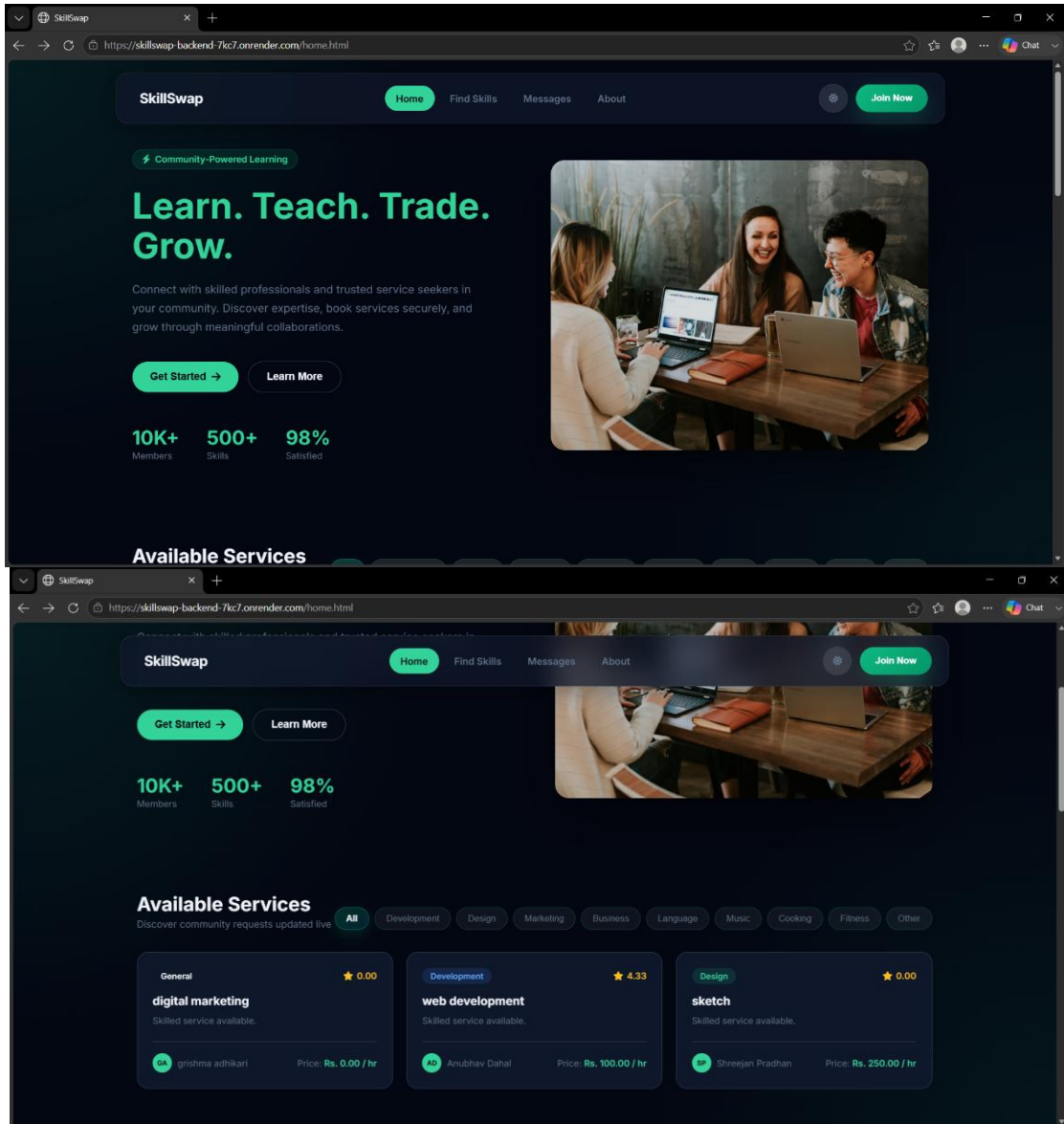
Although the current version of the Skill Swap platform provides the core functionalities required for skill exchange, several enhancements can be implemented in future versions to further improve the system.

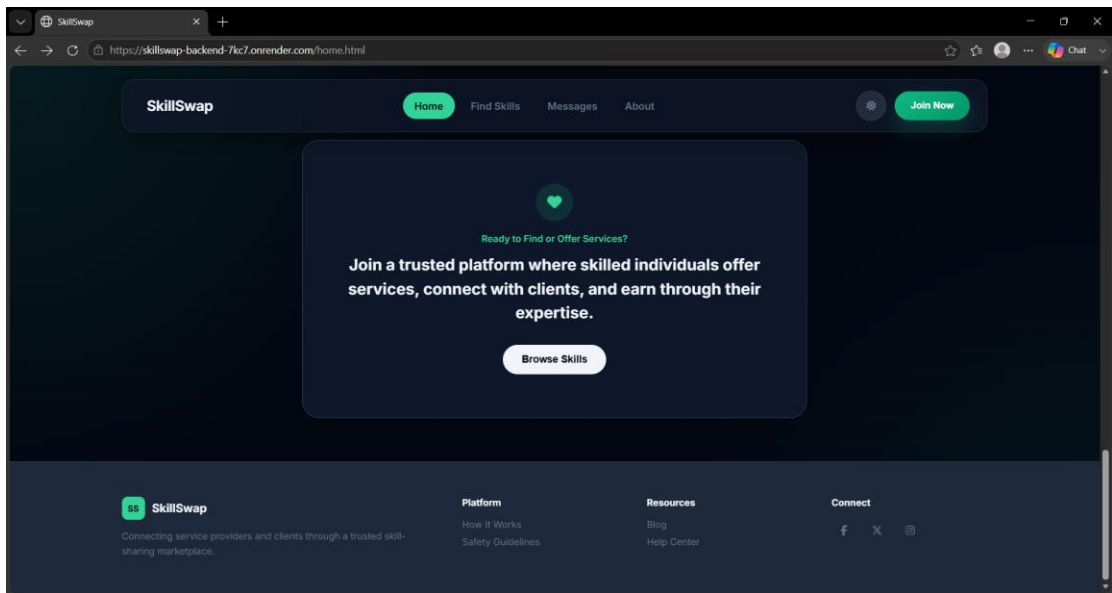
- i. Implement a dedicated mobile application for Android and iOS to improve accessibility and user experience.
- ii. Integrate a secure online payment gateway (such as eSewa, Khalti, or Stripe) to automate online payments instead of relying only on cash or manual QR payment.
- iii. Implement AI-based service recommendations to suggest relevant skills and service providers based on user interests, previous bookings, and search history.
- iv. Add real-time location and map integration using GPS and Google Maps to help users locate nearby service providers and improve service discovery.
- v. Develop an analytics dashboard for administrators to monitor user activity, bookings, revenue, commissions, and overall platform performance through graphical reports.

## APPENDICES

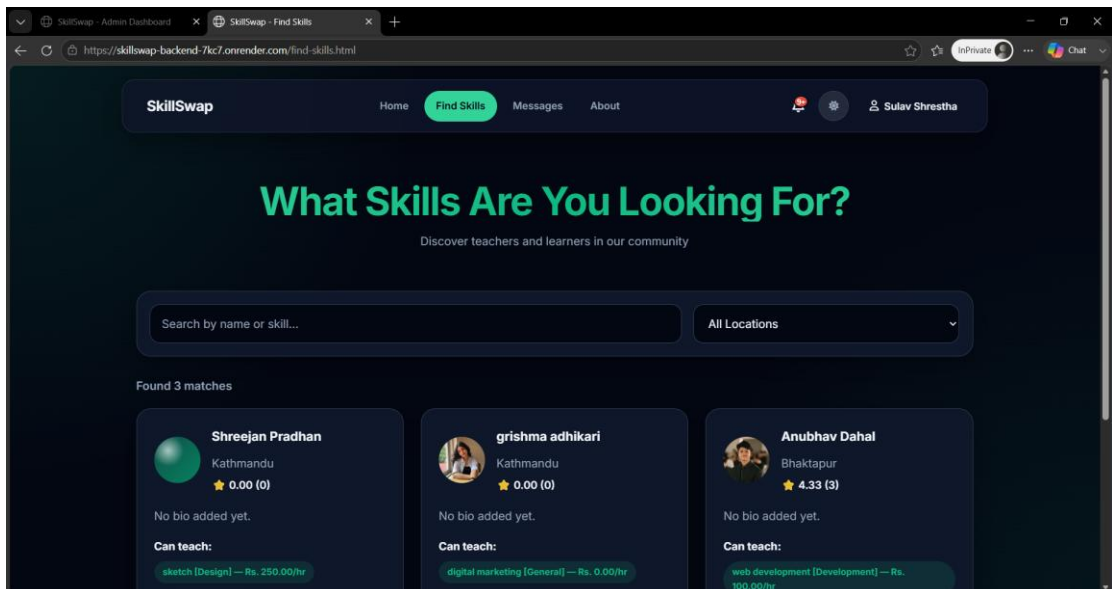
This appendix presents the screenshots of the major interfaces of the Skill Swap system. These interfaces demonstrate the visual design, navigation structure, and functionalities implemented in the application.

### Home Page Interface

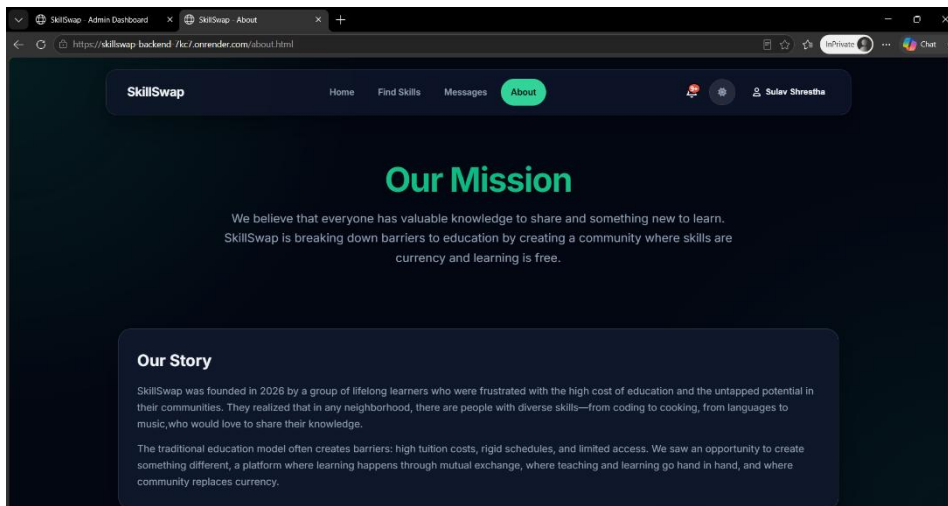


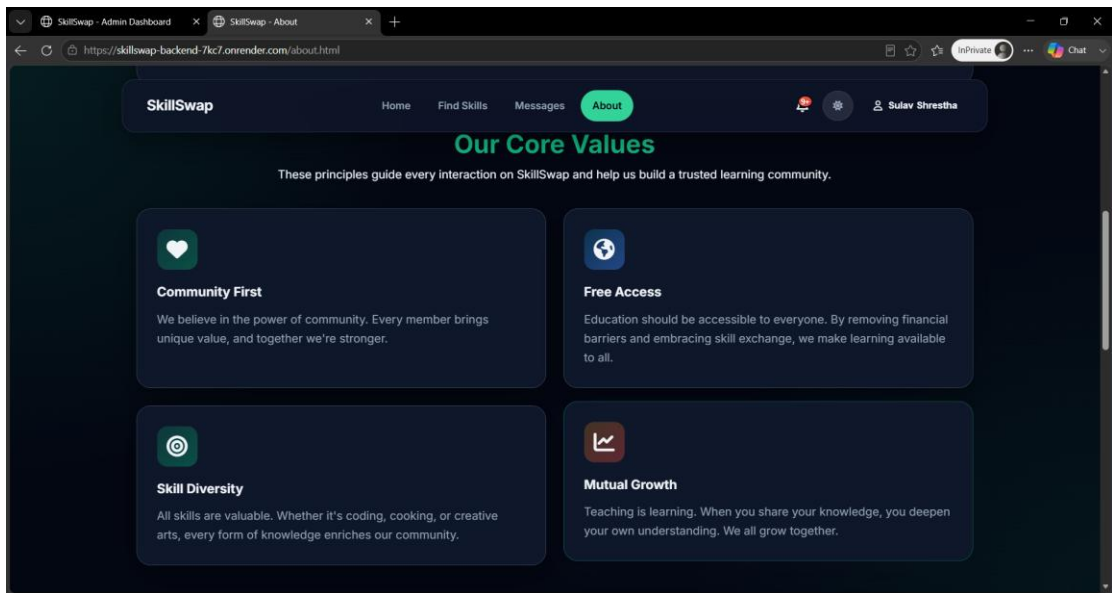


## Find Skills Page Interface

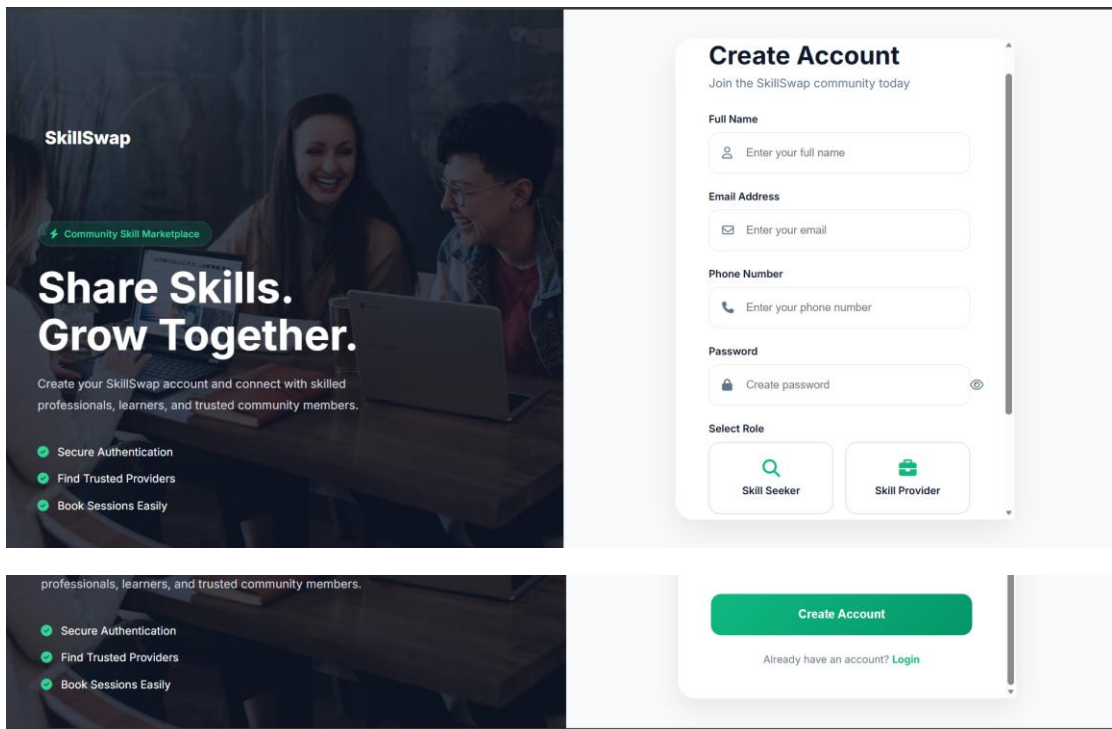


## About Page Interface

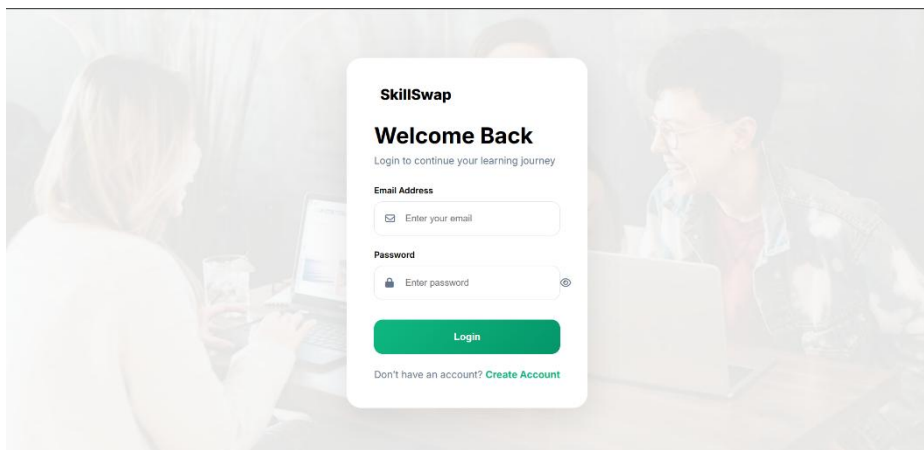




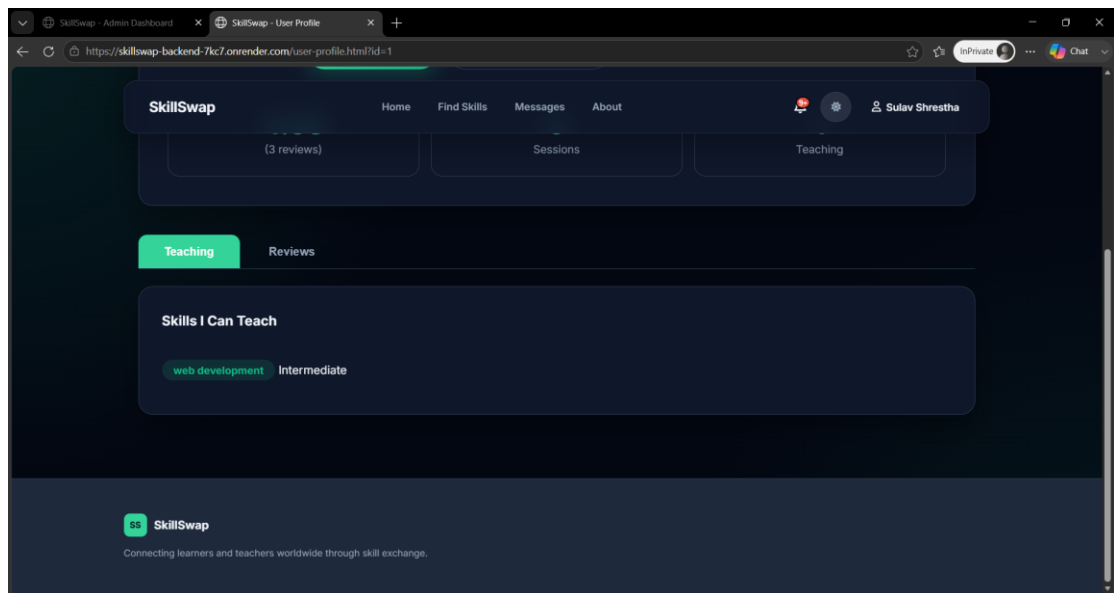
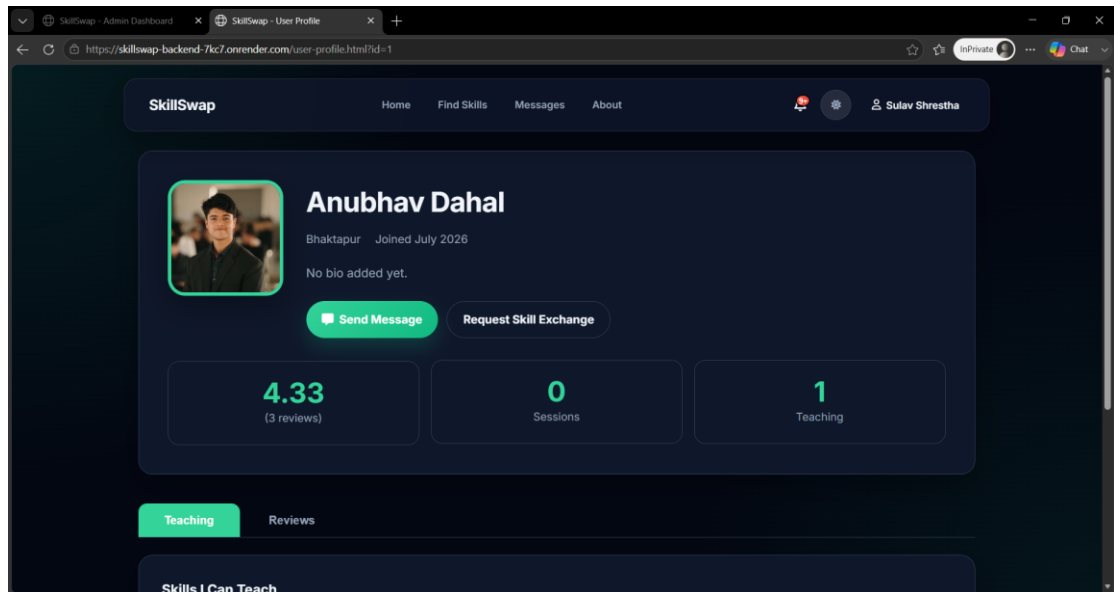
## Sign Up Page Interface



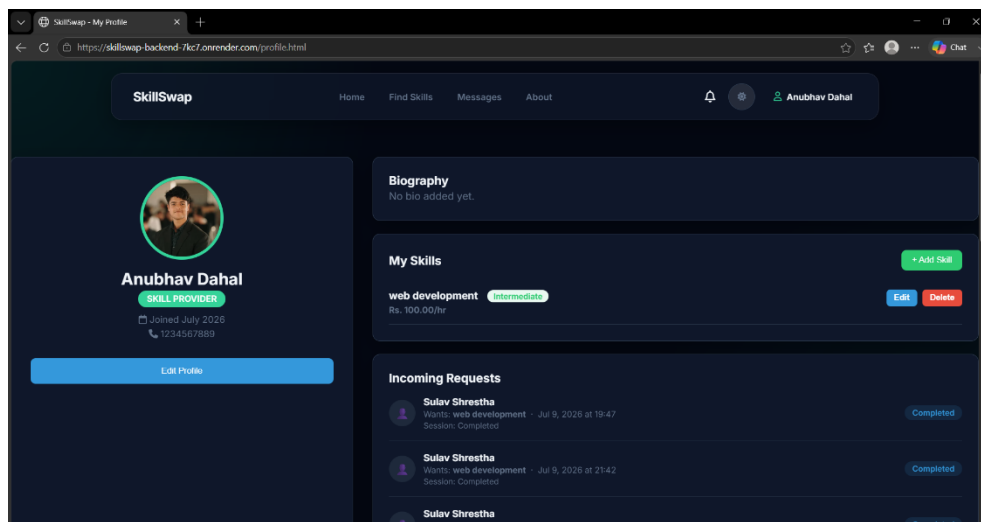
## Login Page Interface



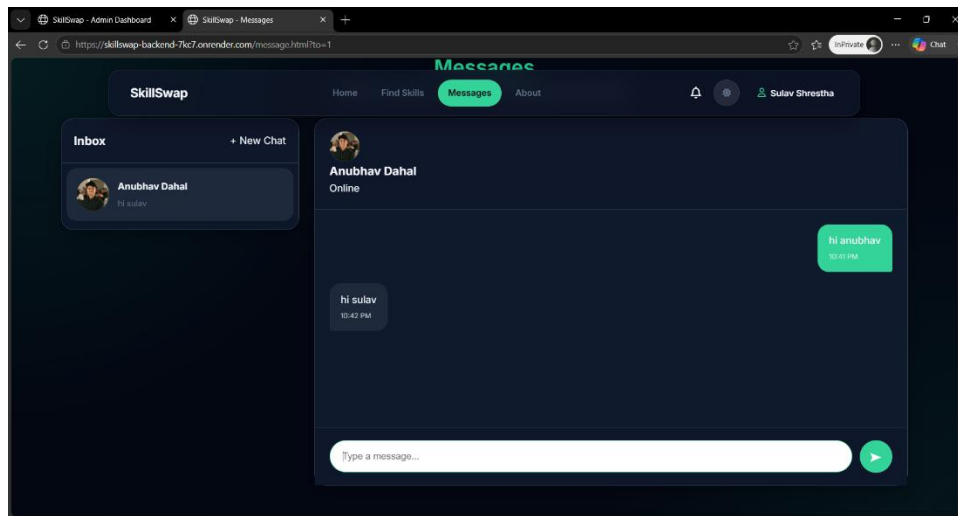
## View Profile Interface



## My Profile Interface



## Message Page Interface





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